

Declare2LTL2FOL4PPM Translation

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1 Primitive FOL vocabulary

Given a prefix l , we define

$$HasAct(l, i, A)$$

meaning that activity A occurs at position i , and

$$Succ(i, j) \equiv i < j \wedge \neg \exists k (i < k < j)$$

for immediate succession.

The predicate

$$Occurs(l, A) \equiv \exists i HasAct(l, i, A)$$

means that activity A occurs at least once somewhere in l .

2 Unary, cardinality, and choice templates

Given a prefix l and an activity A ,

$$AtLeast_n(l, A) \equiv \exists i_1, \dots, i_n ((i_1 < \dots < i_n) \wedge (\bigwedge_{r=1}^n HasAct(l, i_r, A)))$$

means that activity A occurs at least n times in l .

Table 1 lists the unary, cardinality, and choice templates along with their translations into first-order logic (FOL).

Declare Template	LTL Template	FOL definition
Existence(A)	$\Diamond A$	$AtLeast_1(l, A)$
Absence(A)	$\Box \neg A$	$\neg AtLeast_1(l, A)$
AtMost _n (A)	$\neg E_{n+1}(A)$	$\neg AtLeast_{n+1}(l, A)$
Exactly _n (A)	$E_n(A) \wedge \neg E_{n+1}(A)$	$AtLeast_n(l, A) \wedge \neg AtLeast_{n+1}(l, A)$
Init(A)	A	$\exists i (First(l, i) \wedge HasAct(l, i, A))$
End(A)	$\Diamond(A \wedge \neg \bigcirc \top)$	$\exists i (Last(l, i) \wedge HasAct(l, i, A))$
Choice(A,B)	$\Diamond A \vee \Diamond B$	$Occurs(l, A) \vee Occurs(l, B)$
ExclusiveChoice(A,B)	$(\Diamond A \vee \Diamond B) \wedge \neg(\Diamond A \wedge \Diamond B)$	$(Occurs(l, A) \vee Occurs(l, B)) \wedge \neg(Occurs(l, A) \wedge Occurs(l, B))$
RespondedExistence(A,B)	$\Diamond A \rightarrow \Diamond B$	$Occurs(l, A) \rightarrow Occurs(l, B)$
CoExistence(A,B)	$\Diamond A \leftrightarrow \Diamond B$	$Occurs(l, A) \leftrightarrow Occurs(l, B)$
NotCoExistence(A,B)	$\neg(\Diamond A \wedge \Diamond B)$	$\neg(Occurs(l, A) \wedge Occurs(l, B))$

Table 1. Unary, cardinality, and choice templates and the corresponding FOL translations.

3 Response, precedence, succession, alternate, and negative templates

Let $A_i = HasAct(l, i, A)$ and $B_j = HasAct(l, j, B)$, where A and B denotes two activities. Table 2 lists the response, precedence, succession, alternate, and negative templates along with their translations into FOL.

Declare template	LTL template	FOL definition
Response(A,B)	$\Box(A \rightarrow \Diamond B)A$	$\forall i(A_i \rightarrow \exists j(i < j \wedge B_j))$
Precedence(A,B)	$(\neg BU A) \vee \Box(\neg B)$	$\forall j(B_j \rightarrow \exists i(i < j \wedge A_i))$
Succession(A,B)	$Response(A, B) \wedge Precedence(A, B)$	$Response(l, A, B) \wedge Precedence(l, A, B)$
ChainResponse(A,B)	$\Box(A \rightarrow \bigcirc B)$	$\forall i(A_i \rightarrow \exists j(Succ(i, j) \wedge B_j))$
ChainPrecedence(A,B)	$\Box(\bigcirc B \rightarrow A)$	$\forall j(B_j \rightarrow \exists i(Succ(i, j) \wedge A_i))$
ChainSuccession(A,B)	$\Box(A \leftrightarrow \bigcirc B)$	$ChainResponse(l, A, B) \wedge ChainPrecedence(l, A, B)$
AlternateResponse(A,B)	$\Box(A \rightarrow \bigcirc(\neg AU B))$	$\forall i(A_i \rightarrow \exists j(i < j \wedge B_j \wedge \neg \exists k(i < k < j \wedge A_k)))$
AlternatePrecedence(A,B)	$Precedence(A, B) \wedge \Box(B \rightarrow \bigcirc(Precedence(A, B)))$	$\forall j(B_j \rightarrow \exists i(i < j \wedge A_i \wedge \neg \exists k(i < k < j \wedge B_k)))$
AlternateSuccession(A,B)	$AlternateResponse(A, B) \wedge AlternatePrecedence(A, B)$	$AlternateResponse(A, B) \wedge AlternatePrecedence(A, B)$
NotSuccession(A,B)	$\Box(A \rightarrow \neg(\Diamond B))$	$\neg \exists i, j(A_i \wedge B_j \wedge i < j)$
NotChainSuccession(A,B)	$\Box(A \rightarrow \bigcirc(\neg B))$	$\neg \exists i, j(A_i \wedge B_j \wedge \neg Succ(i, j))$

Table 2. Response, precedence, succession, alternate, and negative templates and the corresponding FOL translations.